



Chapter 13: Physical & Facility Security

CompTIA Security+ Study Slides

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Learning Objectives

- Understand **physical and environmental security principles**
- Identify types of **physical access controls** and **surveillance systems**
- Explain **biometric authentication** and error metrics
- Describe **facility protection mechanisms** (locks, barriers, mantraps)
- Discuss **fire suppression, HVAC, and shielding controls**

Overview

Even the most secure network can be compromised if **attackers gain physical access**.

 **Goal:** Protect people, hardware, and data centers from physical threats such as:

- Unauthorized access
- Theft or vandalism
- Natural disasters
- Environmental hazards

Layers of Physical Security

1. **Perimeter Security** – fences, gates, guards, lighting
2. **Building Security** – doors, locks, ID badges, CCTV
3. **Internal Access Control** – mantraps, biometrics, secure zones
4. **Data Center Security** – server cages, access logs, alarms
5. **Environmental Controls** – fire suppression, HVAC, shielding

 *Defense in depth applies to the physical world too.*

Physical Access Controls

Physical Controls: tangible barriers to prevent unauthorized entry.

Control Type	Examples
Mechanical	Locks, keys, safes
Electronic	Smart cards, keypads, RFID badges
Biometric	Fingerprint, iris, facial recognition
Administrative	Security policies, visitor logs, escorts

Security Guards & Access Policies

- Act as deterrents and first responders
- Verify credentials and monitor access points
- Maintain **visitor logs** and **sign-in/out procedures**
- Enforce **two-person control** in high-security areas

 Human oversight complements electronic systems.

CCTV Surveillance

Closed-Circuit Television (CCTV): monitors and records activity for deterrence and forensics.

Type	Description
Fixed Camera	Constant view of a defined area
PTZ (Pan-Tilt-Zoom)	Remote directional and zoom control
Infrared/Night Vision	Records in low-light environments
IP-based CCTV	Digital, networked, supports remote access

Best Practices:

- Retain recordings per policy (e.g., 30–90 days)

- Secure footage against tampering

Door Locks & Entry Systems

Mechanism	Example	Notes
Keyed Locks	Mechanical keys	Simple, inexpensive
Electronic Locks	Keypad PIN, smart card	Centralized management
RFID / Proximity Cards	Badge readers	Common in enterprises
Biometric Locks	Fingerprint, retina scan	High assurance

 Always pair **mechanical + electronic** controls for redundancy.

Mantraps

Mantrap: small vestibule with two interlocking doors that require sequential authentication.

Purpose:

- Prevent tailgating or piggybacking
- Enforce single-person entry
- Often used in data centers and secure areas

 Ensures "**one person, one entry**" policy.

Biometrics

Definition: authentication based on unique physical or behavioral characteristics.

Type	Example
Physiological	Fingerprint, iris, retina, facial recognition
Behavioral	Typing rhythm, voice, gait

Advantages: hard to forge, convenient for users.

Disadvantages: cost, potential privacy concerns, false matches.

12 34 Biometric Accuracy Metrics

Metric	Description
FAR (False Acceptance Rate)	Probability an unauthorized person is accepted
FRR (False Rejection Rate)	Probability an authorized person is denied
CER (Crossover Error Rate)	Point where FAR and FRR are equal; measures overall accuracy

💡 Lower **CER** = higher system accuracy.

Security Zones

Divide facilities into areas with increasing sensitivity.

Zone	Example	Access Level
Public Zone	Reception, lobby	Low
Restricted Zone	Office area	Medium
Secure Zone	Data center, server room	High
Controlled Zone	Vault, command center	Very High

Access Principle: enforce **least privilege** and **need-to-enter**.

Fire Suppression Systems

Purpose: control or extinguish fires without damaging critical assets.

Type	Description	Suitable For
Class A	Ordinary combustibles (wood, paper)	Offices
Class B	Flammable liquids (oil, fuel)	Labs, garages
Class C	Electrical fires	Server rooms
Class D	Combustible metals	Industrial sites
Class K	Kitchen oils and fats	Cafeterias

Fire Suppression Methods

System Type	Mechanism	Notes
Wet Pipe Sprinkler	Water always in pipes	Fast response; risk of water damage
Dry Pipe Sprinkler	Air in pipes until heat triggers	For cold areas or data centers
Pre-Action System	Requires both smoke and heat sensors	Reduces accidental activation
Clean Agent System	Gas-based (FM-200, CO ₂ , Halon)	Safe for electronics

 *Clean agent systems are ideal for server environments.*

HVAC and Environmental Controls

HVAC: Heating, Ventilation, and Air Conditioning

Security Roles:

- Maintain temperature and humidity (~40% humidity ideal)
- Prevent static buildup and overheating
- Filter dust and contaminants
- Connect to building monitoring systems (BMS, SCADA)

 Protect IT equipment from environmental hazards.

Shielding and TEMPEST

Electromagnetic Emanation (EME): unintentional emission of signals that may leak sensitive data.

Countermeasures:

- **STP (Shielded Twisted Pair)** cabling
- **Faraday Cage:** conductive enclosure blocking RF signals
- **TEMPEST Standard:** U.S. gov't specification for emission security
- **EMP Protection:** safeguards against electromagnetic pulses

 Critical for military, intelligence, and financial data centers.

Vehicle & Perimeter Security

- **Barriers/Bollards:** prevent vehicle ramming attacks
- **Guards & Gates:** control vehicle access points
- **License Plate Recognition (LPR)** for authorized vehicles
- **Air Gaps:** physically isolate secure networks from public access

 Combine physical deterrents with procedural controls.

Facility Monitoring & Maintenance

- Regular physical inspections
- Test alarms, sensors, and backup power (UPS/generator)
- Maintain access logs and camera records
- Integrate with **Security Information and Event Management (SIEM)**

 Continuous monitoring = continuous protection.

Best Practices Summary

- Enforce multi-layered physical security
- Use CCTV, mantraps, and access logs for accountability
- Protect against fire, heat, and environmental threats
- Implement clean agent systems for IT areas
- Apply shielding to prevent data leakage

End of Chapter 13

Next: [Chapter 14 — Authentication & Identity Management](#)